

In-class template

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1. Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# reading .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# arrange plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
```

```

aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality
water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))

```

2 . Cleaning

sites object

Goal: create a new object from **sites** that only includes NCOS sites

```

# creating a new object from sites
ncos_sites <- sites |>
  # filtering to only include NCOS quarterly sampling
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

```

clean taxon_list

Goal: convert character strings in the existing **taxon_list** data frame into snakecase

```

taxon_list_clean <- taxon_list |>
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))

```

clean water_quality

Goal: create a data frame with median pH, median DO (mg/L), median salinity (ppt) for each date and site

- create a new object from `water_quality`
- clean column names
- only include surveys from quarterly NCOS sites
- create a new column called `date_site` to join with later data frames
- group by `date_site`
- calculate median pH, salinity, and dissolved oxygen
- ungroup
- filter out any outliers

```
# creating a new object
water_quality_clean <- water_quality |>
  # cleaning column names
  clean_names() |>
  # use a filtering join, only include sites in ncos_sites
  semi_join(ncos_sites,
            by = "site") |>
  # creating new column to join with later data frames
  unite("date_site", date, site,
        remove = TRUE) |>
  # grouping by date/site
  group_by(date_site) |>
  # calculating median, pH, salinity, and DO
  summarize(
    med_ph = median(p_h, na.rm = TRUE),
    med_sal = median(salinity_ppt, na.rm = TRUE),
    med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungrouping data frame
  ungroup() |>
  # filtering out outliers
  filter(date_site != "2022-08-12_NVBR")
```

clean aq_ins

Goal: create a long format data frame in which each row contains the counted individuals in each taxon during each survey. The data frame should include taxonomic information for each organism and the water quality information for each survey

- create a new object from `aq_ins`
- clean column names
- filter to only include quarterly NCOS sites
- select columns
- filter to only include “filtered beaker” sample
- convert data frame to long format

- group by site, date and taxon
- calculate average abundance per liter (sum counts divided by 7.5)
- create a new column called `date_site` to join with water quality
- join data frames
- join with taxon list

```
aq_ins_clean <- aq_ins |>
  clean_names() |>
  semi_join(ncos_sites,
            by = "site") |>
  rename(date = date_on_vial) |>
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  filter(sample_type %in% c("FB250", "FB 250")) |>
  pivot_longer(
    # columns to make longer
    cols = ostracod:annelida_polychaete,
    # new columns for the old column names
    names_to = "taxon",
    # new column for the values in each of the old columns
    values_to = "abundance"
  ) |>
  group_by(date, site, taxon) |>
  summarise(ave_lit = sum(abundance, na.rm = TRUE) / 7.5) |>
  ungroup() |>
  unite("date_site", date, site, remove = FALSE) |>
  left_join(water_quality_clean, by = "date_site") |>
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelp Bridge",
    site == "NPB2" ~ "Phelps Road",
```

```
site == "NWP" ~ "West Pond",  
site == "NDC" ~ "Devereux Creek"  
)
```